Lecture 14

Collapse and branching: GRW and Everett

Objectives. *GRW changes universal unitary dynamics; Everett changes the requirement of one global outcome. We compute the exact and approximate claims behind a GRW localisation, and prove that relative branches reproduce L10's full reading-chain probabilities and conditional states. A common comparison then returns to the course's question: what data specify a quantum system, and which physical commitments remain open?*

- Derive GRW hit weights and a quantitative one-hit selection estimate, distinguishing exact expectations from approximate selection.
- Compute the labelled rigid-configuration damping rate and prove the finite reading-chain relative-state identity, including mixed inputs.
- Compare the three responses on equal questions and specify states, algebras, dynamics, composites and instruments consistently with superselection.

14.1 GRW: stochastic localisations without a meter

In the one-dimensional model, a wavefunction evolves by its Hamiltonian between randomly timed hits. Hits occur at Poisson rate λ_{GRW} and have operators

$$L_z = (\pi r_C^2)^{-1/4} e^{-(Q-z)^2/(2r_C^2)}, \qquad r_C > 0. \tag{14.1}$$

The continuous label z is a localisation centre, not an angular-momentum component. Gaussian integration gives $\int \|L_z\psi\|^2\, dz = \|\psi\|^2$. Indeed this is exactly L10's Gaussian pointer family with $g = 1$ and meter width $r_C/\sqrt{2}$, whose normalisation was proved there. The identity $\int L_z^\dagger L_z dz = \mathbb{1}$ means its positive scalar pairings, not an operator-norm integral.

GRW postulates the centre density $q(z) = \|L_z\psi\|^2$ and the selected state $L_z\psi/\sqrt{q(z)}$ where $q(z) > 0$, without a physical apparatus being present. Equality to a mathematical instrument family does not turn this stochastic law into a unitary measurement of an unobserved meter. Its nonunitary random hits are the change to U.

Proposition 14.1 (Exact branch weights). *Let $\psi = \sum_{j=1}^m \sqrt{p_j}\psi_j$, with unit branches having disjoint position supports, $p_j \geq 0$ and $\sum_j p_j = 1$. Put $a_j(z) = \left\|L_z\psi_j\right\|^2$. Expectations and probabilities below use the centre density $q(z)dz$. A hit gives branch weights*

$$w_j(z) = \frac{p_j a_j(z)}{\sum_k p_k a_k(z)}, \qquad \mathbb{E}_z[w_j] = p_j.$$

Proof. Multiplication by L_z preserves the support orthogonality, so $q(z) = \sum_j p_j a_j(z)$. The jth squared norm after normalising by q is the stated w_j. Averaging with the *centre density* gives $\int q(z) w_j(z) dz = p_j \int a_j(z) dz = p_j$. ∎

The post-hit branch shape is $L_z\psi_j/\sqrt{a_j(z)}$, not generally ψ_j. Preserving expected weights does not mean selecting one branch: when the packets are unresolved, the weights can remain nearly unchanged. Here is the additional separation argument.

Suppose the finite branch supports lie in $[x_j - \ell, x_j + \ell]$, with centre separations at least d and $\ell < d/3$. Define disjoint centre regions $R_j = [x_j - d/3, x_j + d/3]$. Explicitly,

$$a_j(z) = \int |\psi_j(x)|^2 \frac{e^{-(z-x)^2/r_C^2}}{\sqrt{\pi} r_C}\, dx.$$

Conditional on the jth density a_j, the centre has the scalar distribution of a position in that packet plus Gaussian noise of density $e^{-u^2/r_C^2}/(\sqrt{\pi} r_C)$. Hence

$$\delta_j := \int_{R_j^c} a_j(z) dz \le \operatorname{erfc}\left(\frac{d/3 - \ell}{r_C}\right), \qquad \delta := \sum_j p_j \delta_j \le \operatorname{erfc}\left(\frac{d/3 - \ell}{r_C}\right).$$

Here $\operatorname{erfc}(b) = 2\int_b^\infty e^{-u^2} du/\sqrt{\pi}$ is just the Gaussian two-sided tail. These bounds follow from the support restriction: leaving R_j requires noise of magnitude at least $d/3 - \ell$. For $0 < \epsilon < 1/2$ set $S_j = \{z \in R_j : w_j(z) \ge 1 - \epsilon\}$. Then

$$p_j - \delta - \delta/\epsilon \le \Pr(S_j) \le p_j + \delta, \qquad \Pr\left(\bigcup_j S_j\right) \ge 1 - \delta - \delta/\epsilon. \tag{14.2}$$

To prove the estimates, since $R_j \subset R_k^c$ for $k \ne j$ and $q(1 - w_j) = \sum_{k \ne j} p_k a_k$, leakage from all wrong packet densities into R_j is at most δ, so $p_j - \delta \le \Pr(R_j) \le p_j + \delta$. Also $\int_{R_j} q(1 - w_j) dz$ is exactly that wrong-packet leakage. The probability in R_j where $1 - w_j > \epsilon$ is at most this integral divided by ϵ. Summing over disjoint R_j gives total leakage at most δ, while centres outside their union have probability at most δ. These observations give both bounds in (14.2).

When the support gap satisfies $(d/3 - \ell)/r_C \gg 1$, $\delta \to 0$: one branch has weight near one with high probability and its label probabilities approach p_j. No lower bound on the prior weights was assumed. The stronger sufficient condition $\ell \ll r_C \ll d$ also keeps a selected narrow packet nearly undistorted for typical hit centres; narrowness is not necessary for the weight-selection estimate alone. The exact statement requires disjoint support; overlapping wave-packet tails require additional errors. Since a Gaussian multiplier is strictly positive, a finite hit does not exactly delete every other nonzero branch. The treatment of those small tails and of physical records also involves the model's ontology.

Self-check 14.1

1. Why does $\mathbb{E}[w_j] = p_j$ alone not imply selection?
2. Which error in (14.2) grows if packets become unresolved?

14.2 The averaged channel and amplification

The unread one-hit map $\mathcal{T}(\rho) = \int L_z \rho L_z dz$ is L10's unread Gaussian pointer channel. Completing the square in its kernel gives

$$\mathcal{T}(\rho)(x, x') = e^{-(x-x')^2/(4r_C^2)} \rho(x, x').$$

Equivalently it is L11's proved random-unitary construction with kick density $r_C e^{-r_C^2 k^2}/\sqrt{\pi}$. The identification follows from the same Gaussian Fourier integral, with the kernel/pairing scope proved in L10–L11. With free motion neglected the Poisson average is

$$\Phi_t(\rho) = e^{-\lambda_{\mathrm{GRW}} t} \sum_{n=0}^{\infty} \frac{(\lambda_{\mathrm{GRW}} t)^n}{n!} \mathcal{T}^n(\rho).$$

The nonnegative coefficients sum to one. Positive trace tails construct the sum, and the same construction after every finite spectator proves CP, as in L7. Thus it is a channel. Its kernel multiplier is the scalar exponential series

$$\exp\left[-\lambda_{\mathrm{GRW}} t \left(1 - e^{-(x-x')^2/(4r_C^2)}\right)\right]. \tag{14.3}$$

It saturates at rate λ_{GRW} for resolved separations. It is not L11's purely quadratic diffusion kernel. Including a Hamiltonian gives the ensemble equation

$$\dot{\rho} = -\frac{i}{\hbar}[H, \rho] + \lambda_{\mathrm{GRW}}(\mathcal{T}(\rho) - \rho).$$

This full equation is *sketched*: unbounded Hamiltonian domains require additional work. The zero-free-motion channel and its kernel above were constructed directly, without invoking an unproved unbounded GKSL theorem.

For three spatial dimensions replace the prefactor in (14.1) by $(\pi r_C^2)^{-3/4}$ and the squared distance by $|Q - z|^2$. For N *labelled* constituents with independent rate λ_{GRW} per label, each hit changes only its coordinate. Multiplying their independent Poisson kernel factors gives damping rate

$$\Gamma(q, q') = \lambda_{\mathrm{GRW}} \sum_{i=1}^{N} \left(1 - e^{-|q_i - q_i'|^2/(4r_C^2)}\right). \tag{14.4}$$

Evaluating this continuous coefficient at matched rigid translations $q_i' = q_i + \mathbf{d}$ gives $N\lambda_{\mathrm{GRW}}(1 - e^{-d^2/(4r_C^2)})$, $d = |\mathbf{d}|$. Point configurations label the coefficient, not Hilbert-space states; using it for coherence between narrow normalised packets is an approximation. Deformations or rotations require the full sum of different displacements.

With conventional model choices $\lambda_{\mathrm{GRW}} = 10^{-16}\,\mathrm{s}^{-1}$, $r_C = 10^{-7}\,\mathrm{m}$, and a pointer with $N = 10^{23}$ labels at separation $d = 10^{-6}\,\mathrm{m}$, the total hit rate is $10^7\,\mathrm{s}^{-1}$ and the overlap factor is e^{-25}. The first-hit waiting time is about $10^{-7}\,\mathrm{s}$; it is also approximately the ensemble coherence time here. To apply the selection estimate in three dimensions, use for each label the slabs perpendicular to $\mathbf{d}$ in place of R_j. The Gaussian noise component along $\mathbf{d}$ has the same one-dimensional density, so the leakage proof applies to the full centre probabilities and global branch weights. For two rigid branches whose constituent parallel support half-width is $\ell = 0.01 r_C$, the present $d = 10 r_C$ gives $\delta \le \mathrm{erfc}(10/3 - 0.01) \simeq 2.60 \cdot 10^{-6}$. With $\epsilon = 0.01$, some branch has weight at least 0.99 with probability at least $1 - 101\delta > 0.9997$, after one hit, on the stated $10^{-7}\,\mathrm{s}$ mean first-hit scale. Compact support idealises real tails, which would add an error term. For $d \lesssim r_C$, the rate is smaller and a single hit need not resolve the alternatives.

The physics

The parameters are conventional hypotheses, not measured constants. Bassi et al., *RMP* **85**, 471 (2013), review model-dependent experimental constraints as of that review. GRW predicts extra coherence loss and momentum noise even when known

environmental noise is removed; its non-dissipative kicks also inject mean kinetic energy. Calling that a temperature rise needs an equilibration and heat-capacity model. Environmental scattering can exchange energy too. Isolated-system residual noise and matter-wave visibility, rather than the word "heating" alone, provide tests. L11's air-grain estimate concerns a different object and reservoir, not a numerical measurement of this pointer's collapse rate. The labelled calculation is a simplifying model: individual labelled hits need not preserve an identical-fermion subspace. Symmetry-preserving collapse models require further specification. GRWm's matter density and GRWf's flashes are distinct choices of physical ontology for records.

Self-check 14.2

1. Expand the rate for $d \ll r_C$. Why is it not the total hit rate?
2. Does $N\lambda_{\mathrm{GRW}}$ by itself establish first-hit selection?

14.3 Everett: relative states reproduce the reading chain

Everett retains unitary dynamics and treats outcomes as relative to branches, rather than D's unique global outcome. We prove the particular agreement required by the course: L10's probabilities and conditional states are already encoded in a unitary chain with retained records. The proof is finite dimensional, and does not assert a unique fundamental branch count.

Let the first instrument have finite Kraus operators $K_{j\alpha}$, the second $M_{k\beta}$, and let U be the intermediate system unitary. Both Kraus lists have total completeness $\mathbb{1}$. L7's unitary dilations use fresh independently prepared pure probes, with orthogonal output labels $|j,\alpha\rangle$ and $|k,\beta\rangle$. These labelled probes include the dilation registers. The second coupling acts on the system and its fresh probe, leaving the first record untouched. Define the composed operation $\mathcal{J}_{jk} = \mathcal{I}'_k \circ \mathrm{Ad}_U \circ \mathcal{I}_j$, where $\mathrm{Ad}_U(X) = UXU^\dagger$.

Theorem 14.2 (Finite chain identity). *Immediately after these two readings, the global pure state has orthogonal history components Ψ_{jk} with weights $w_{jk} = \|\Psi_{jk}\|^2 = \mathrm{Tr}\,\mathcal{J}_{jk}(\rho)$. For positive w_{jk} its reduced relative system state is $\mathcal{J}_{jk}(\rho)/w_{jk}$. If ρ is mixed, purify it with a reference R; the partial trace defining that system state includes R as well as both probes and all dilation registers.*

Proof. For pure input ψ the unnormalised history vector is

$$\Psi_{jk} = \sum_{\alpha,\beta} M_{k\beta} U K_{j\alpha} \psi \otimes |j,\alpha\rangle \otimes |k,\beta\rangle\,.$$

The total is $\sum_{jk} \Psi_{jk}$. Probe orthogonality gives $w_{jk} = \sum_{\alpha,\beta} \left\|M_{k\beta} U K_{j\alpha} \psi\right\|^2 = \mathrm{Tr}\,\mathcal{J}_{jk}(|\psi\rangle\langle\psi|)$. Tracing the probes removes their cross terms and leaves exactly

$$\sum_{\alpha,\beta} M_{k\beta} U K_{j\alpha}\, |\psi\rangle\langle\psi|\, K_{j\alpha}^\dagger U^\dagger M_{k\beta}^\dagger.$$

This is $\mathcal{J}_{jk}(|\psi\rangle\langle\psi|)$, proving the claim. For mixed input take a purification $\Omega \in \mathcal{H}_S \otimes \mathcal{H}_R$. Apply each Kraus product tensored with $\mathbb{1}_R$ to Ω in the displayed sum. Its squared norm is the same trace by the purification pairing. The partial trace over R and both probes yields $\mathcal{J}_{jk}(\rho)$, by that pairing or a finite spectral expansion. Tracing only the probes would

instead leave an operator on $S + R$. Normalise only when $w_{jk} > 0$. Completeness gives $\sum_{jk} w_{jk} = 1$. ∎

For L10's actual capstone there is one Kraus operator per outcome: $K_j = E_j$ and the second operator is M_k. The four weights are $5/16, 3/16, 3/16, 5/16$, and the relative system vectors are exactly its four tabulated vectors. Replacing the first instrument by $K_j = |+x\rangle\langle j|$ instead gives $(4 - k\sqrt{3})/16$ and the reset final states. The theorem therefore reproduces both full sequences, not just ideal repeatable measurements.

Let Π_h project onto each history's orthogonal record sector. For any bounded A commuting with every Π_h, cross terms vanish and

$$\langle \Psi, A\Psi \rangle = \sum_{h:w_h>0} w_h \left\langle \Psi_h/\sqrt{w_h}, A\Psi_h/\sqrt{w_h} \right\rangle .$$

A subsequent unitary commuting with these record projections preserves the weights, though it can evolve the within-branch states. The theorem's conditional-state identity refers to the end of the specified chain; further dynamics must be included in the composed operation if compared. Actual macroscopic decoherence gives approximately robust records, not a uniquely specified exact branching decomposition for every situation.

A record-erasing unitary is outside that record-preserving premise. L12's inverse CNOTs restore system σ_x mean one from the coherent three-qubit state and zero from its incoherent mixture. Unitarity cannot merge the orthogonal total states: the inverse CNOTs send $|000\rangle$ to $|000\rangle$ and $|111\rangle$ to $|100\rangle$. The record factors reset while the two system alternatives stay orthogonal, so the incoherent mixture retains zero σ_x mean. Every surviving external copy must also be coherently controlled to recover that interference.

Self-check 14.3

1. Why must the purification reference be traced out for a mixed input?
2. Do retained weights imply that subsequent within-branch states stop evolving?

14.4 Weights, probabilities and the same comparison questions

The identity uses squared norms and the instrument probabilities already defined by Born's rule. It does not derive the interpretation of those weights as probabilities from unitary dynamics alone, or select one global branch. Wallace's decision-theoretic programme is a named representation argument with additional assumptions: sufficiently rich available acts, coherent preferences across decisions, continuity, preferences depending only on the resulting state (state supervenience), macrostate indifference, and indifference to branch splitting that leaves rewards unchanged (§5.6 pp. 174–181; theorem discussion §5.7 p. 182 onward). Its conclusion is reported under those axioms, not as a proof from U alone. Counting branches is not supplied as an alternative probability law.

Locality also needs the model specified. GRW does not inherit PI merely from calling its wavefunction complete. For separate noninteracting labs, averaging a remote hit gives a local TP map; its centre probability cancels the conditional normalisation, leaving the other lab's marginal unchanged (cited: Tumulka, §5.5.9 pp. 232–234). At the predictive wavefunction-state level this gives PI under the stated setting assumptions; Bell-violating correlations then require OI to fail. Conditioning on future realised collapse centres would define a different fine description. The selected wavefunction changes nonlocally even when averaged marginals do not signal. Everett likewise retains the Bell correlations; rejecting a

unique global outcome is not construction of a classical factorising single-outcome model (Wallace, §8.7 p. 310).

Question	Bohm	GRW	Everett
State	Wavefunction plus configuration	Stochastic wavefunction	Unitary state with relative records
Ontology	Configuration and wavefunction	GRWm matter density or GRWf flashes	Universal wavefunction
Dynamics	Schrödinger and guiding flow	Unitary intervals and random hits	Universal unitary dynamics
Records	Actual apparatus configuration	Localisation with small tails; GRWm/f account	Branch-relative records
Probabilities	Quantum equilibrium, equivariant	Postulated centre law	Norm weights; interpretation needs further argument
Locality	Fine deterministic PI fails; equilibrium nonsignalling	Noise-averaged nonsignalling; conditional changes nonlocal	Usual Bell correlations; no unique-global-outcome model assumed
Tests	Equilibrium quantum predictions	Additional isolated noise and coherence loss	Standard unitary predictions
Main change	C	U	D

The table compares the stated models, not every possible modification. GRW's record ontology and residual tails need their own account; a large weight is the proved approximate selection criterion, not exact deletion. Bohmian nonequilibrium variants and modified unitary theories would have additional empirical claims beyond this comparison.

The physics

Other approaches make different commitments. Copenhagen accounts specify a classical measurement context for applying the quantum rules. QBism interprets state probabilities as an agent's commitments, with Born's rule a consistency constraint, rather than a microscopic collapse mechanism (Fuchs–Schack, *RMP* **85**, 1693 (2013)). Consistent histories assigns classical joint probabilities only within an appropriate consistent history family; its single-framework rule forbids freely combining incompatible families (Griffiths, Ch. 16 pp. 216–219; the histories construction is in Ch. 10 pp. 137–147). These are stated positions, not conclusions selected by our no-go proof.

14.5 The quantum-system specification

Definition 14.3 (The course's system card)**.** A quantum-system specification gives a nonzero separable Hilbert space, its admitted observable algebra and normal states, allowed dynamics and symmetries, the composite rule, and the instruments used for observations.

For the full algebra $\mathcal{B}(\mathcal{H})$, states are exactly L1's density operators and pure states are L2's rays. With the two-sector algebra of L5, states instead have the unique block-diagonal representatives proved there; its pure states are single-sector rays. Unbounded observables must have spectral projections in the admitted algebra. State and purity are relative to that algebra, not to a choice of Hilbert-space representative alone.

Allowed symmetries must respect this specification. Within a coherent sector Wigner supplies unitary or antiunitary ray implementations. Sector-preserving ones act within

sectors; more general algebra-preserving discrete symmetries may permute equivalent sectors. For example, swapping two isomorphic blocks sends $A_1 \oplus A_2$ to $A_2 \oplus A_1$ and preserves the block algebra without fixing its sector projections. Which such maps are physical symmetries is additional data. The baseline closed evolution is a sector-compatible unitary group under L5's continuity and lift premises; GRW deliberately changes that dynamical law.

Composites use L6's Hilbert tensor product. Their observable algebra must also be specified: imposing each part's sector restriction and imposing only a total conserved charge are different choices. For systems without superselection the course used the full composite algebra. A part initially prepared independently of its environment has the channel description proved in L7; the finite semigroup model has L8's GKSL structure.

For a worked L5-type card, take $\mathcal{H} = \mathbb{C} \oplus \mathbb{C}^2$, $\mathcal{A} = \mathbb{C} \oplus M_2$, $\rho = 1/3 \oplus (2/3)\,|+x\rangle\langle+x|$ and $H = 0 \oplus (\hbar\nu/2)\sigma_z$. Time evolution is the corresponding block unitary. The sector instrument $\mathcal{I}_j(\rho) = E_j \rho E_j$ has effects $E_j \in \mathcal{A}$, probabilities $1/3, 2/3$, and admissible conditional states 1 and $|+x\rangle\langle+x|$ at this initial time. With an ordinary qubit partner, choose composite algebra $M_2 \oplus M_4$ on the two composite sectors. This specifies the algebra as well as the tensor space.

Effects alone do not ensure that an operation descends to admitted states. On the simpler algebra $\mathbb{C} \oplus \mathbb{C}$, the ray $|+x\rangle\langle+x|$ and $\mathbb{1}_2/2$ represent the same restricted state. The one-outcome channel with $K_0 = |0\rangle\langle+x|$, $K_1 = |1\rangle\langle-x|$ has total effect $\mathbb{1}$ in that algebra, yet sends those representatives to $|0\rangle\langle0|$ and $\mathbb{1}_2/2$. Their later sector probabilities differ. Hence an instrument must act unambiguously on the admitted states, with admissible outputs, not merely have effects in the algebra. Operations defined directly on block-diagonal states avoid that representative dependence; permitted sector transfers are additional physical data, not a requirement that each Kraus operator commute with the sector sign.

This common operational structure leaves L12's C/U/D question open. Bohm adds configurations; GRW changes dynamics and specifies record ontology; Everett changes the account of global versus relative records. The card states what each proposal must specify and what its predictions can be compared against, without declaring one interpretation a theorem.

Self-check 14.5

1. For two one-dimensional sectors with energies $\varepsilon_0, \varepsilon_1$, write a state, a compatible Hamiltonian and the sector-reading instrument.
2. Does an algebra-preserving discrete symmetry necessarily fix each sector?

What is a quantum system, and what remains open?

The observable algebra and its normal states specify expectations and events; tensor products specify composites, compatible dynamics specify change, and instruments specify observations. On the full algebra the states are density operators; superselection changes their identification and purity. The formalism predicts complete reading-chain statistics, but universal unitarity, completeness and one global outcome cannot all hold in L12's calibrated model; which premise to change is a further physical or interpretive choice. The three responses were compared by the same mathematical and experimental questions.

Summary. GRW's expected branch weights are exact, whereas one-hit selection needs separation and a tail estimate. Its Poisson jump kernel gives labelled amplification, with explicit modelling limits. Everett's finite branch weights and reduced relative states repro-

duce the full capstone, including a mixed input's reference trace. That identity leaves the probability interpretation as a further argument. The system card keeps states, algebras, dynamics and instruments together and makes the remaining commitments explicit.

Self-check answers.

14.1 1. Weights can remain near their original values on every hit. 2. The Gaussian leakage δ, and hence δ/ϵ.

14.2 1. $N\lambda_{\mathrm{GRW}}d^2/(4r_C^2)$; unresolved alternatives have strongly overlapping hit responses. 2. No; the support gap relative to r_C determines resolution. A narrow packet also suffers less shape change at a typical hit.

14.3 1. Otherwise the result is a state on system plus reference, not the system conditional state. 2. No; record-preserving dynamics can change states within each branch.

14.5 1. $\rho = \mathrm{diag}(w, 1-w)$, $H = \mathrm{diag}(\varepsilon_0, \varepsilon_1)$, and $\mathcal{I}_j(\rho) = E_j \rho E_j$, with $0 \le w \le 1$. 2. No; a permitted swap of equivalent blocks preserves the algebra.

Sources. Tumulka, *Foundations of Quantum Mechanics*, pp. 104–109, 124–129, 232–234 (his width convention has $r_C = \sqrt{2}\sigma$; our GRW/Bassi choice $r_C = 10^{-7}$ m corresponds to $\sigma \simeq 7.1 \cdot 10^{-8}$ m); Ghirardi–Rimini–Weber, *PRD* **34**, 470 (1986); Bassi et al., *RMP* **85**, 471 (2013). Wallace, *The Emergent Multiverse*, pp. 94, 174–182, 308–311; Griffiths, *Consistent Quantum Theory*, pp. 137–147, 216–219. The explicit tail bound and finite capstone identity are supplied here.